
Sprint review #12

HyDE feed recommendation

API Spec Link :

Orchestrator :

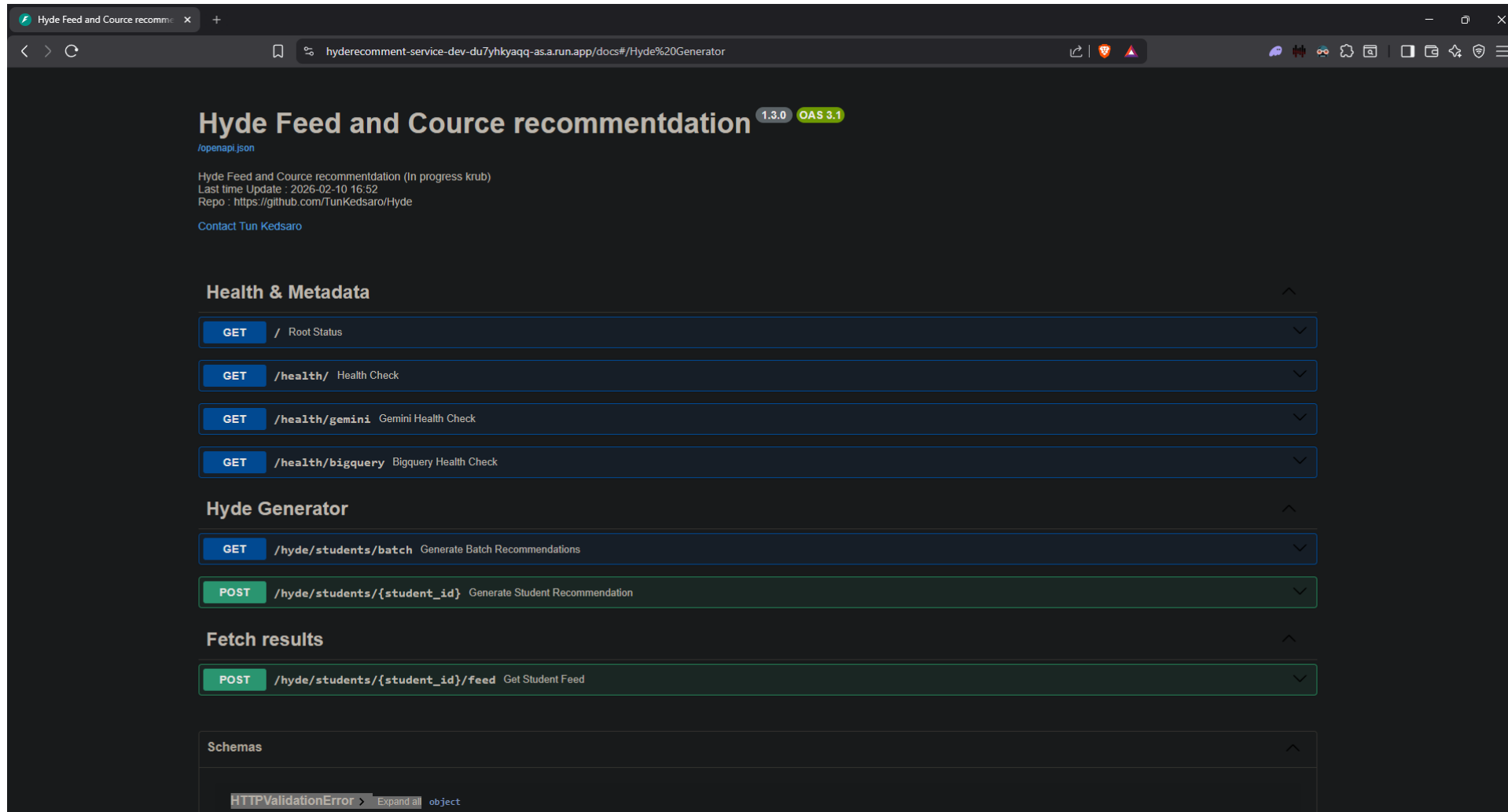
API core evaluation link : <https://hyderecomment-service-du7yhkyaqq-as.a.run.app>

Github : <https://github.com/TunKedsaro/Hyde>

QA : <https://humanintelligence.atlassian.net/wiki/spaces/CPM/pages/134611027/QA+Test+Plan+CV+Evaluation+Pipeline>

Test : <https://humanintelligence.atlassian.net/wiki/spaces/CPM/pages/148636743/HyDE+Core+Recommendation+Engine+Test+Plan>

HyDE Feed Recommendation – HyDE generation



HyDE recommendation core engine API

GCS / hyde-datalake (Test)

HyDE Feed Recommendation – HyDE generation

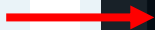
Overview

```
{
  "student_id": "stu_p001",
  "status": "\"hyde-datalake /\n|- stu_p001 /\n|- metadata folder /\n|- metadata.json /\n|- hyde folder /\n|- stu_p001/hyde/hyde_text01.txt /\n|- stu_p001/hyde/hyde_text02.txt /\n|- stu_p001/hyde/hyde_text03.txt x\n|- stu_p001/hyde/hyde_text04.txt /\n|- stu_p001/hyde/hyde_text05.txt /\n|- embedding folder /\n|- stu_p001/embedding/embedding01.npy /\n|- stu_p001/embedding/embedding02.npy /\n|- stu_p001/embedding/embedding03.npy x\n|- stu_p001/embedding/embedding04.npy /\n|- stu_p001/embedding/embedding05.npy /\n\"",
  "metadata": {
    "student_id": "stu_p001",
    "current_status": "student3yr",
    "education_level": "bachelor",
    "education_major": "วิทยาการคอมพิวเตอร์",
    "target_roles": "Data Analyst",
    "timezone": "UTC",
    "model_name": "gemini-2.5-flash",
    "max_output_tokens": 2048,
    "feed_text_max_chars": 240,
    "temperature": 0.2
  },
  "hyde": {
    "hyde_context1": "โลเคชันโปรแกรมทำพอร์ต Data Analyst ด้วย Python SQL",
    "hyde_context2": "เทคนิคเตรียมสัมภาษณ์งาน Data Analyst โจทย์ SQL Python",
    "hyde_context3": "",
    "hyde_context4": "เครื่องมือสร้าง Dashboard วิเคราะห์ข้อมูลออนไลน์",
    "hyde_context5": "ทักษะการสื่อสารนำเสนอข้อมูลสำหรับ Data Analyst"
  },
  "embedded_vector": {
    "embedding_vector1": [
      0.023251675069332123,
      -0.014571324922144413,
      0.01352651696652174
    ],
    "embedding_vector2": [
      0.007405612617731094,
      -0.03062768094241619,
      0.0034155375324189663
    ],
    "embedding_vector3": [],
    "embedding_vector4": [
      -0.011106484569609165,
      0.014421403408050537,
      -0.0009799925610423088
    ],
    "embedding_vector5": [
      0.0059158275835216045,
      -0.007175480481237173,
      0.007349800784140825
    ]
  }
}
```

```
hyde-datalake /
|- stu_p001 /
  |- metadata folder /
    |- metadata.json /
  |- hyde folder /
    |- stu_p001/hyde/hyde_text01.txt /
    |- stu_p001/hyde/hyde_text02.txt /
    |- stu_p001/hyde/hyde_text03.txt x
    |- stu_p001/hyde/hyde_text04.txt /
    |- stu_p001/hyde/hyde_text05.txt /
  |- embedding folder /
    |- stu_p001/embedding/embedding01.npy /
    |- stu_p001/embedding/embedding02.npy /
    |- stu_p001/embedding/embedding03.npy x
    |- stu_p001/embedding/embedding04.npy /
    |- stu_p001/embedding/embedding05.npy /
```

HyDE Feed Recommendation – HyDE generation

```
{
  "student_id": "stu_p001",
  "status": ""hyde-datalake-feeds /\n|- stu_p001 /\n  |- metadata folder /\n    |- metadata.json /\n  |- hyde folder /\n    |- stu_p001/hyde/hyde_text01.txt x\n    |- stu_p001/hyde/hyde_text02.txt x\n    |- stu_p001/hyde/hyde_text03.txt x\n    |- stu_p001/hyde/hyde_text04.txt x\n  """,
  "metadata": {
    "student_id": "stu_p001",
    ...,
    "interaction": [
      {
        "user_id": "stu_p001",
        "feed_id": "TH_F008",
        "ts": "2026-01-04 14:05:02+00:00",
        "event_type": "click",
        "dwell_ms": "0"
      },
      ...
    ],
    "hyde": {
      "hq": [
        {
          "query_id": "Q1",
          "query_text": "แนวทางสร้างพอร์ด Data Analyst โปรแกรม Python SQL",
          "weight": 1,
          "intent_label": "history_aligned"
        },
        ...
        {
          "query_id": "Q5",
          "query_text": "เทคนิคการนำเสนอข้อมูลและเล่าเรื่องด้วย Data Storytelling",
          "weight": 0.6,
          "intent_label": "exploratory"
        }
      ]
    },
    "embedded_vector": {
      "embedding_vector1": [
        0.018321901559829712,
        -0.008828291669487953,
        ...,
        0.012498840689659119
      ],
      ...,
      "embedding_vector5": [
        0.0003302044642623514,
        0.019586067646741867,
        ...,
        -0.015933198854327202
      ]
    }
  }
}
```



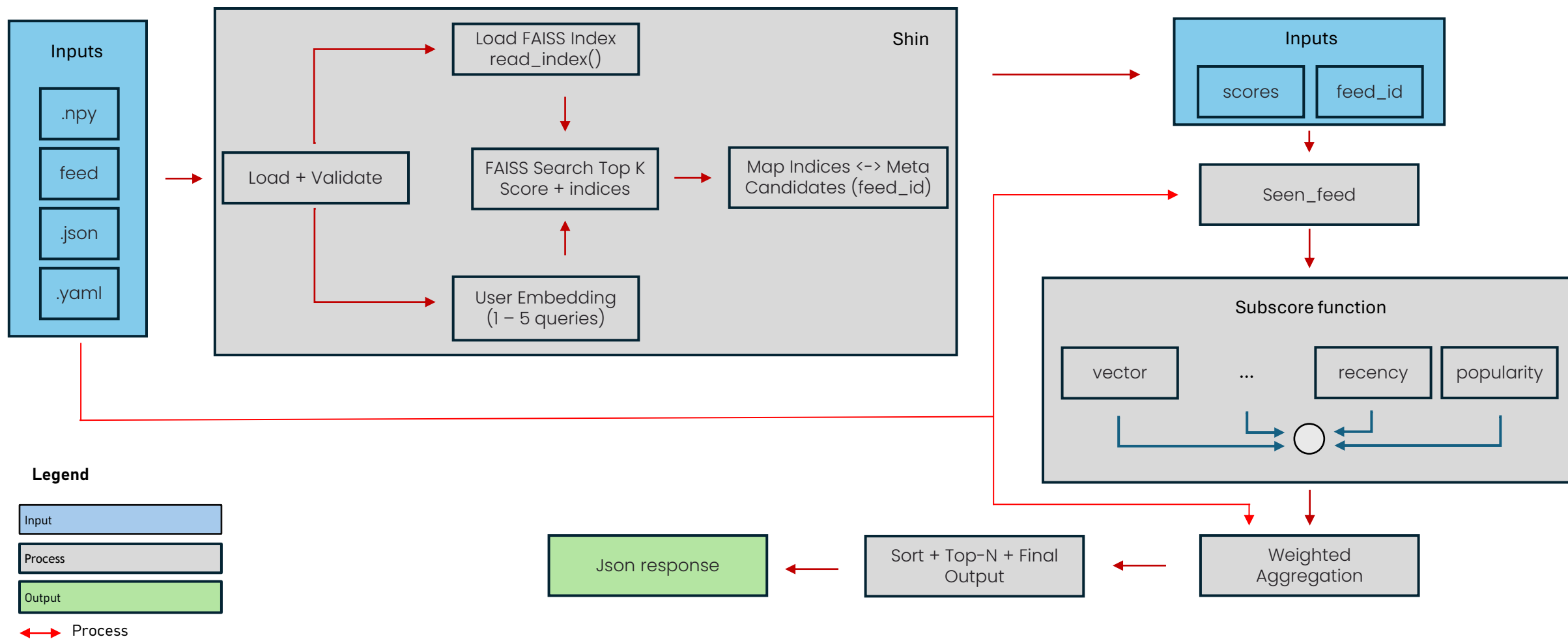
```
hyde-datalake-feeds
├── stu_p001
│   ├── embedding folder
│   │   ├── stu_p001/embedding/embedding01.npy
│   │   ├── stu_p001/embedding/embedding02.npy
│   │   ├── stu_p001/embedding/embedding03.npy
│   │   ├── stu_p001/embedding/embedding04.npy
│   │   └── stu_p001/embedding/embedding05.npy
│   ├── hyde folder
│   │   └── stu_p001/hyde/hyde.json
│   └── metadata folder
│       └── stu_p001/metadata/metadata.json
```

HyDE Feed Recommendation – HyDE generation

```
{
  "student_id": "stu_p001",
  "status": ""hyde-datalake-feeds /\n|- stu_p001 /\n |- metadata folder /\n |- metadata.json /\n |- hyde folder /\n |- stu_p001/hyde/hyde_text01.txt x\n |- stu_p001/hyde/hyde_text02.txt x\n |- stu_p001/hyde/hyde_text03.txt x\n |- stu_p001/hyde/hyde_text04.txt x\n """,
  "metadata": {
    "student_id": "stu_p001",
    ...,
    "interaction": [
      {
        "user_id": "stu_p001",
        "feed_id": "TH_F008",
        "ts": "2026-01-04 14:05:02+00:00",
        "event_type": "click",
        "dwell_ms": "0"
      },
      ...
    ]
  },
  "hyde": {
    "hq": [
      {
        "query_id": "Q1",
        "query_text": "แนวทางสร้างพอร์ด Data Analyst โปรแกรม Python SQL",
        "weight": 1,
        "intent_label": "history_aligned"
      },
      ...
      {
        "query_id": "Q5",
        "query_text": "เทคนิคการนำเสนอข้อมูลและเล่าเรื่องด้วย Data Storytelling",
        "weight": 0.6,
        "intent_label": "exploratory"
      }
    ]
  },
  "embedded_vector": {
    "embedding_vector1": [
      0.018321901559829712,
      -0.008828291669487953,
      ...
      0.012498840689659119
    ],
    ...
    "embedding_vector5": [
      0.0003302044642623514,
      0.019586067646741867,
      ...
      -0.015933198854327202
    ]
  }
}
```

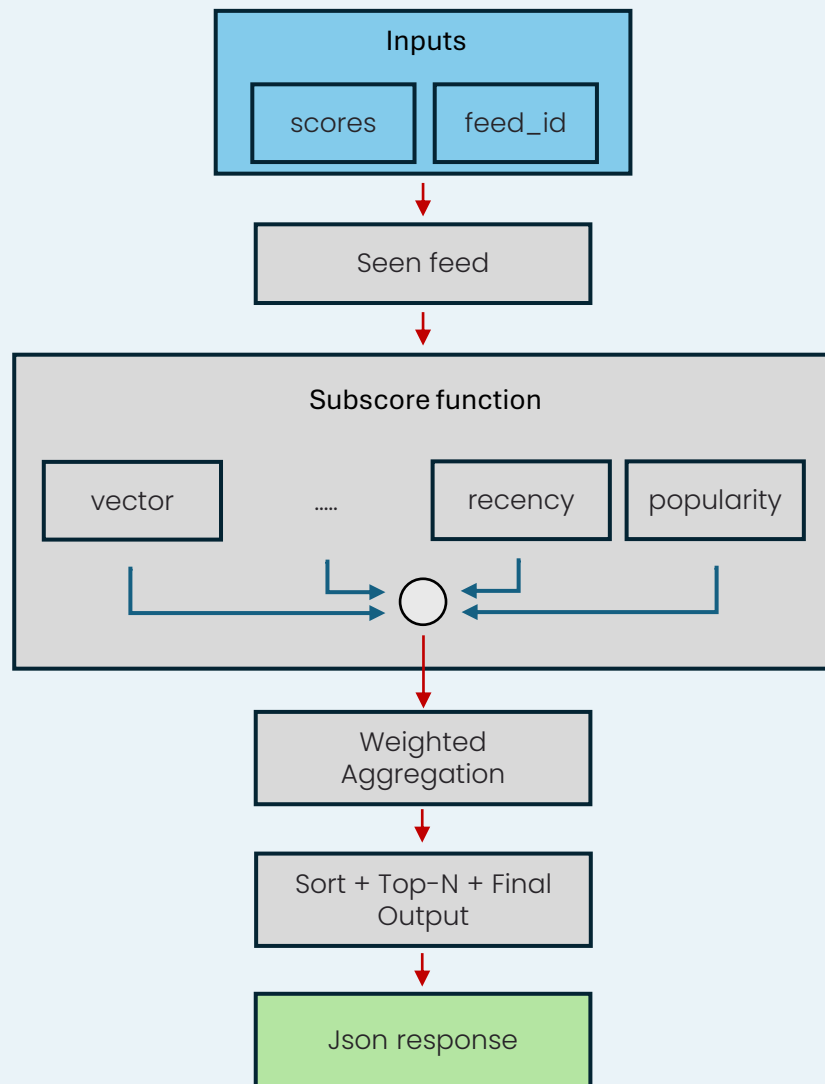
```
{
  "user_id": "stu_p001",
  "feed_id": "TH_F008",
  "ts": "2026-01-04 14:05:02+00:00",
  "event_type": "click",
  "dwell_ms": "0"
},
{
  "user_id": "stu_p001",
  "feed_id": "TH_F001",
  "ts": "2026-01-06 13:14:30+00:00",
  "event_type": "comment",
  "dwell_ms": "0"
},
...
{
  "user_id": "stu_p001",
  "feed_id": "TH_F001",
  "ts": "2026-01-06 13:12:10+00:00",
  "event_type": "view",
  "dwell_ms": "52000"
}
```

HyDE Feed Recommendation - Subscore



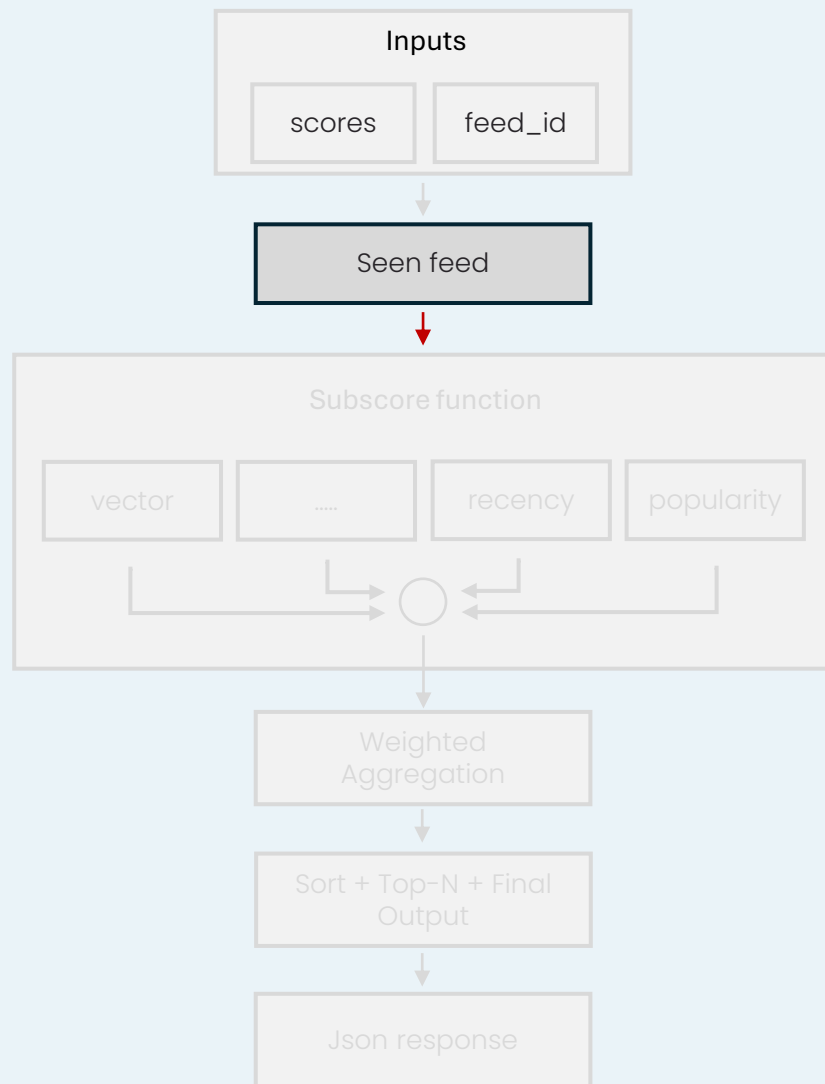
HyDE Feed Recommendation - Subscore

Overview



HyDE Feed Recommendation - Subscore

Overview

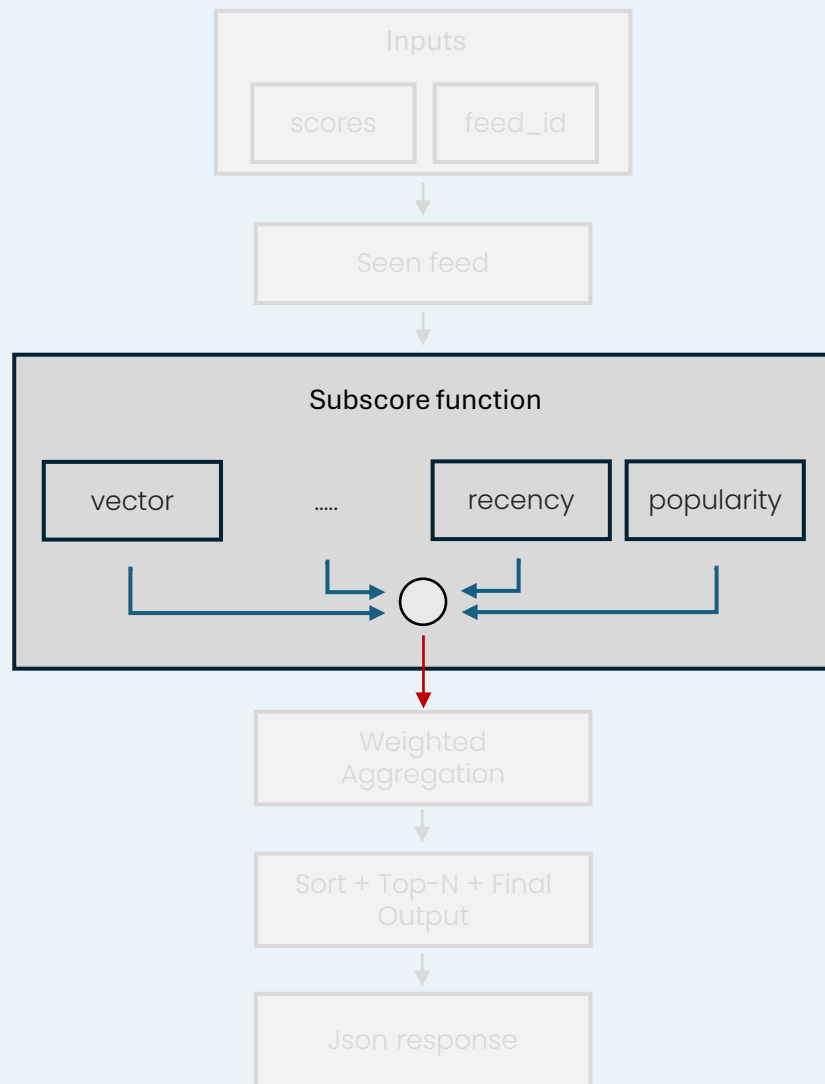


```
"metadata": {
  "student_id": "stu_p001",
  "current_status": "student",
  ...,
  "temperature": 0.2,
  "interaction": [
    {
      "user_id": "stu_p001",
      "feed_id": "TH_F001",
      "ts": "2026-01-06 13:13:05+00:00",
      "event_type": "like",
      "dwell_ms": "0"
    },
    ...,
    {
      "user_id": "stu_p001",
      "feed_id": "TH_UNI_043",
      "ts": "2026-01-06 06:40:00+00:00",
      "event_type": "view",
      "dwell_ms": "12000"
    },
    ...
    {
      "user_id": "stu_p001",
      "feed_id": "TH_F003",
      "ts": "2026-01-05 09:20:11+00:00",
      "event_type": "view",
      "dwell_ms": "41000"
    }
  ]
}
```

```
Feed set (11) -> ['EN_F027', 'EN_F028', 'EN_F029', 'TH_BIO_056', 'TH_BIO_058', 'TH_F002', 'TH_F004', 'TH_F005', 'TH_F001', 'TH_UNI_043', 'TH_F003']
Seen set (3) -> {'TH_F001', 'TH_F003', 'TH_UNI_043'}
Never seen (8) -> {'EN_F029', 'TH_BIO_058', 'EN_F028', 'TH_F002', 'TH_BIO_056', 'TH_F004', 'TH_F005', 'EN_F027'}
```


HyDE Feed Recommendation - Subscore

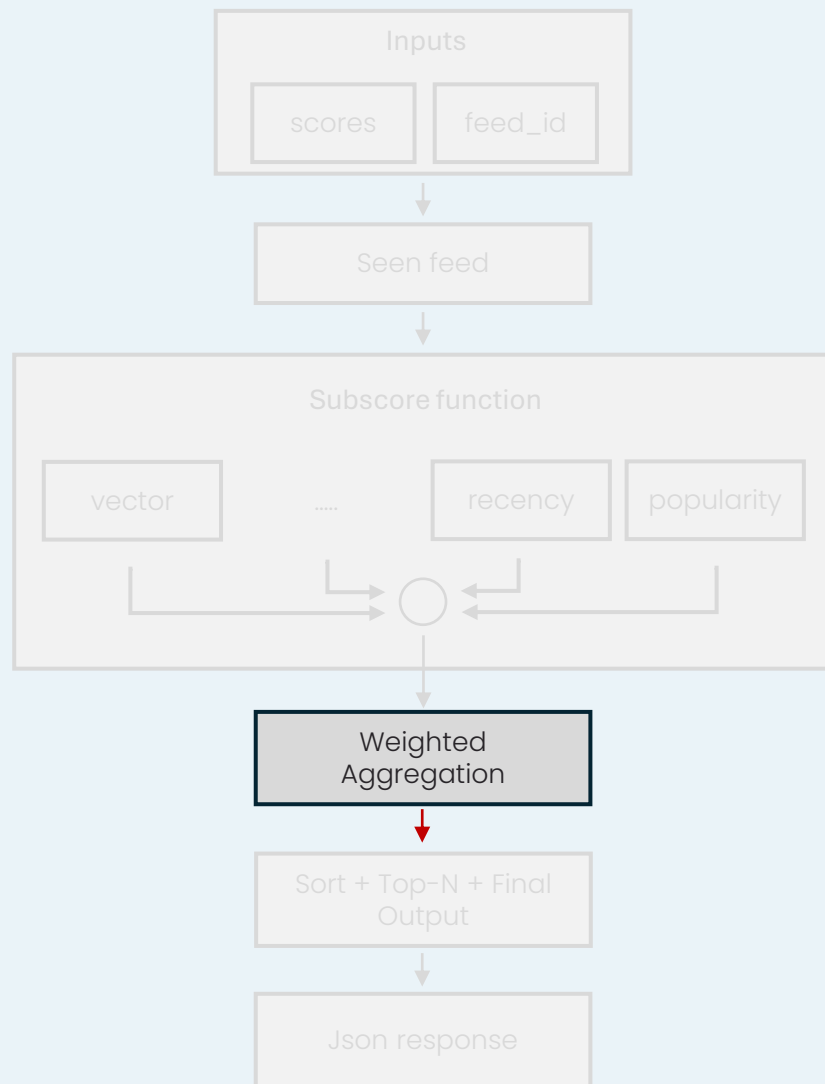
Overview



```
"candidates": [  
  {  
    "feed_id": "EN_F028",  
    "vector_score": 0.8667604923248291,  
    "subscores": {  
      "language_match": 0.0,  
      "recency": 0.19370910075390038,  
      "popularity": 0.7434854245159717  
    },  
    "feed_header": "How to Build a Strong Data Analytics Portfolio",  
  },  
  {  
    "feed_id": "TH_F005",  
    "vector_score": 0.8074958324432373,  
    "subscores": {  
      "language_match": 1.0,  
      "recency": 0.2555596828044196,  
      "popularity": 0.7719089301210578  
    },  
    "feed_header": "Roadmap Data Analyst.... ทำอะไร ก่อนสมัครงาน",  
  },  
  {  
    "feed_id": "EN_F027",  
    "vector_score": 0.8040957450866699,  
    "subscores": {  
      "language_match": 0.0,  
      "recency": 0.21215839045810933,  
      "popularity": 0.7501787727241187  
    },  
    "feed_header": "SQL Interview Questions for Entry-Level Data Analysts",  
  },  
]
```

HyDE Feed Recommendation - Subscore

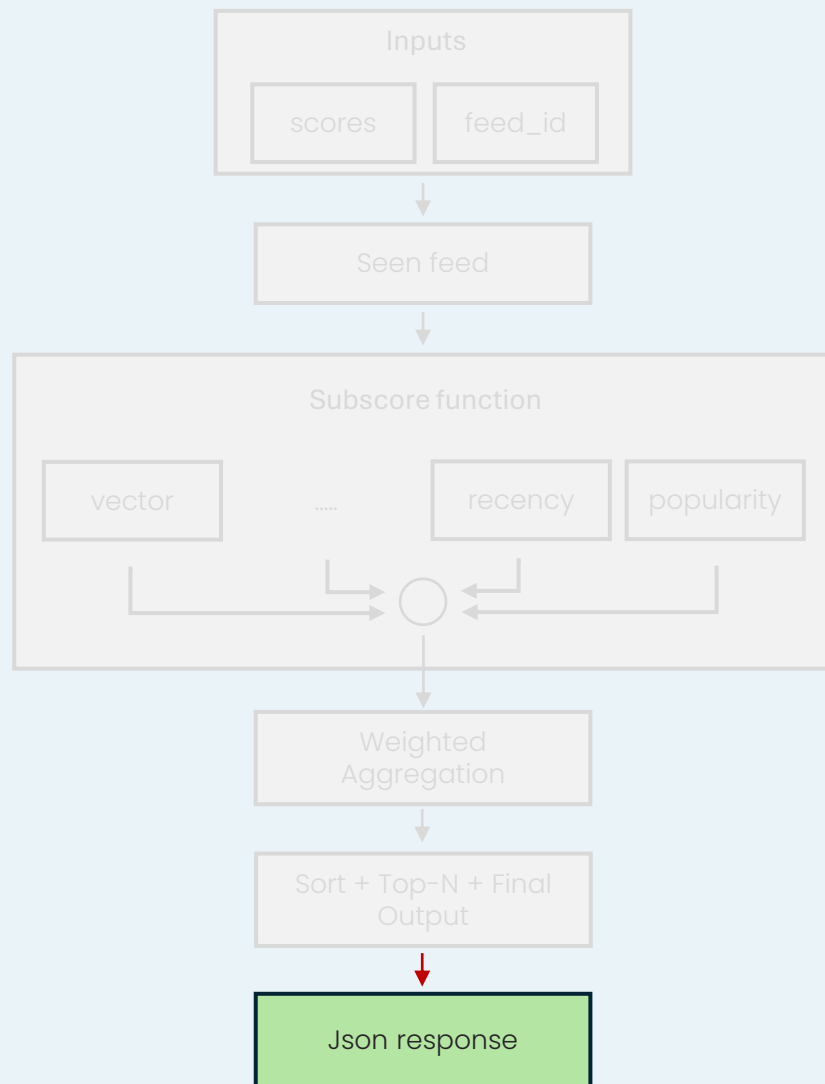
Overview



```
"candidates": [
  {
    "feed_id": "EN_F028",
    "vector_score": 0.8667604923,
    "subscores": {
      "language_match": 0.0,
      "recency": 0.19370,
      "popularity": 0.74348
    },
    "feed_header": "How to Build a Strong Data Analytics Portfolio",
    "final_score": 0.66767
  },
  {
    "feed_id": "TH_F005",
    "vector_score": 0.8074958324432373,
    "subscores": {
      "language_match": 1.0,
      "recency": 0.25555,
      "popularity": 0.77190
    },
    "feed_header": "Roadmap Data Analyst ... ทำอะไร ก่อนสมัครงาน",
    "final_score": 0.66061
  },
  {
    "feed_id": "EN_F027",
    "vector_score": 0.8040957450866699,
    "subscores": {
      "language_match": 0.0,
      "recency": 0.21215,
      "popularity": 0.75017
    },
    "feed_header": "SQL Interview Questions for Entry-Level Data Analysts",
    "final_score": 0.64263
  }
],
"weights": {
  "vector_score": 0.5,
  "language_match": 0.0,
  "recency": 0.25,
  "popularity": 0.25
},
```

HyDE Feed Recommendation - Subscore

Overview



```
"ranked": [  
  {"feed_id": "EN_F028", "final_score": 0.6676788774798825},  
  {"feed_id": "TH_F005", "final_score": 0.660615069452988},  
  {"feed_id": "EN_F027", "final_score": 0.642632163338892},  
  {"feed_id": "EN_F029", "final_score": 0.63919870533132},  
  {"feed_id": "TH_F013", "final_score": 0.6369022489921211},  
  {"feed_id": "TH_F015", "final_score": 0.6307819786983692},  
  {"feed_id": "TH_F026", "final_score": 0.6213201494315035},  
  {"feed_id": "TH_F021", "final_score": 0.6173446742539355}  
]
```

HyDE Feed Recommendation - Subscore

Test ID	Description	Method	Type	Result	Evidence
5.1.1	Scorer (user_state, candidates) E2E contract	PyTest	Contract	PASS	test_5_1_scorer_contract.py
5.1.2	return_debug=False production shape	PyTest	Contract	PASS	test_5_1_scorer_contract.py
5.1.3	Deterministic same input → same output * Decimal issue	PyTest	Determinism	PASS	test_5_1_scorer_contract.py
5.2.1	Popularity log formula correct	PyTest	Unit	PASS	test_5_2_popularity.py
5.2.2	Recency exponential decay correct	PyTest	Unit	PASS	test_5_2_recency.py
5.2.3	YAML weight change deterministic	PyTest	Aggregation	PASS	test_5_2_yaml_determinism.py
5.3.1	Tie-breaker stable	PyTest	Ranking	PASS	test_5_3_tie_breakers.py
5.3.2	Post-processing preserves order	PyTest	Ranking	PASS	test_5_3_post_processing.py
5.3.3	NaN safety	PyTest	Robustness	PASS	test_5_3_nan_safety.py

HyDE Feed Recommendation - Subscore

Run	Top1	Top2	Top3	Top4	Top5
1	EN_F028 (0.66477)	TH_F005 (0.656778)	EN_F027 (0.639446)	EN_F029 (0.636669)	TH_F013 (0.634252)
2	EN_F028 (0.66477)	TH_F005 (0.656777)	EN_F027 (0.639446)	EN_F029 (0.636669)	TH_F013 (0.634252)
3	EN_F028 (0.66477)	TH_F005 (0.656777)	EN_F027 (0.639446)	EN_F029 (0.636669)	TH_F013 (0.634252)
4	EN_F028 (0.664769)	TH_F005 (0.656777)	EN_F027 (0.639446)	EN_F029 (0.636668)	TH_F013 (0.634252)
...					
10	EN_F028 (0.664768)	TH_F005 (0.656775)	EN_F027 (0.639444)	EN_F029 (0.636667)	TH_F013 (0.634252)
delta	-0.000002	-0.000003	-0.000002	-0.000002	0.000000

Delta (Δ) = ± 0.000003

```
"ranked": [  
  {  
    "feed_id": "EN_F028",  
    "final_score": 0.664768  
  },  
  {  
    "feed_id": "TH_F005",  
    "final_score": 0.656775  
  },  
  {  
    "feed_id": "EN_F027",  
    "final_score": 0.639444  
  },  
  {  
    "feed_id": "EN_F029",  
    "final_score": 0.636667  
  },  
  {  
    "feed_id": "TH_F013",  
    "final_score": 0.63425  
  },  
  {  
    "feed_id": "TH_F015",  
    "final_score": 0.628358  
  },  
  {  
    "feed_id": "TH_F026",  
    "final_score": 0.619865  
  },  
  {  
    "feed_id": "TH_F021",  
    "final_score": 0.615509  
  }  
]
```

HyDE Feed Recommendation - Subscore

Run	load_params	extract_query weight	Vector retrieval	Exclude seen	Build candidate	Score aggregation	total_ms
1	2.857	0.008	0.431	13.244	0.135	2.012	183.546
2	3.023	0.024	0.211	4.087	0.076	1.399	126.233
3	3.002	0.009	0.109	3.91	0.092	1.994	122.945
4	2.933	0.009	0.113	4.01	0.101	1.527	119.943
5	3.013	0.008	0.133	4.478	0.1	2.244	128.047
6	5.121	0.019	0.241	8.278	0.1	1.248	126.97
7	5.414	0.022	0.243	8.259	0.078	1.093	133.769
8	4.277	0.021	0.241	7.845	0.098	1.375	131.184
9	2.941	0.009	0.113	3.998	0.1	1.521	132.694
10	2.893	0.009	0.116	4.051	0.087	1.346	123.491
avg	3.5474	0.0138	0.1951	6.216	0.0967	1.5759	132.8822